



## M002 Week 2 — Turning (Advanced)

**Topic:** Turn at Walls · **Course:** Maze Madness · **Level:** Advanced · **Time:** about 55 minutes

The rule: wall ahead, turn. No wall, move forward. This week you write the real code.

### 1 · Trace It 🧠

```
while True:
    if agent.detect(BLOCK, FORWARD):
        agent.turn(RIGHT_TURN)
    else:
        agent.move(FORWARD, 1)
```

Say what the agent does each loop.

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## 2 · Read the API

| Code                                      | Means          |
|-------------------------------------------|----------------|
| <code>agent.detect(BLOCK, FORWARD)</code> | wall in front? |
| <code>agent.turn(RIGHT_TURN)</code>       | turn right     |
| <code>agent.turn(LEFT_TURN)</code>        | turn left      |
| <code>agent.move(FORWARD, 1)</code>       | go forward 1   |

**Write one line: turn left if a wall is on the LEFT.**

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### 3 · Spot the Bug 🐛

```
# A
if agent.detect(BLOCK, FORWARD):
    agent.turn(RIGHT_TURN)
    agent.move(FORWARD, 1)
```

**What is wrong in A?**

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```
# B
if agent.detect(BLOCK, LEFT):
    agent.turn(RIGHT_TURN)
else:
    agent.move(FORWARD, 1)
```

**B should turn at a front wall. What is wrong?**

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4 · Fix It 

Rewrite B so it turns left when a wall is in front, else moves forward.

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## 5 · Write Full Code

Loop forever. Wall ahead? Turn right. No wall? Move forward 1.

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## 6 · Chat Command

`player.on_chat("go", fn)` runs `fn` when you type `go` in chat.

**Write code so typing `go` starts your maze loop.**

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## 7 · Finish the Maze 📷

Open your homework world. Get the agent out with your turn code.

Take a photo or video at the maze end.

**Phone narration plan — say each line out loud while you show the run:**

The maze turns where ...

I checked `agent.detect(BLOCK, ...)` for ...

When a wall was ahead I ...

My agent got stuck when ...

I fixed it by ...

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Send this worksheet + your walkthrough video to teacher on KakaoTalk.